

INCIDENT INVESTIGATION REPORT

Proxy Telemetry Analysis — HaveIBeenPwned / VirusTotal / zone-h.org Activity

Classification	Internal — Confidential
Case Reference	INC-2021-1116-01
Report Type	DFIR — Log-Based Findings
Evidence Source	Blue Coat ProxySG Access Logs
Subject Account	Fahad3315 (IP: 10.55.66.777)
Analysis Period	2021-11-16, 09:52–11:42 (UTC+3)
Investigator	Aljoharah Alqahtani — Cybersecurity Analyst
Prepared For	Cybersecurity IR Team & Investigation Committee
Report Status	Final — Log-Source-Limited

1. Executive Summary

Proxy telemetry for user Fahad3315 on 2021-11-16 reveals an active security-verification pattern between 10:03 and 11:42 (UTC+3): repeated queries to VirusTotal and HaveIBeenPwned (HIBP), interleaved with a perimeter-blocked attempt to access a known defacement repository (zone-h.org).

This telemetry establishes an observable timeline connecting the account to the broader incident window. Because the dataset is restricted to outbound web-proxy logs, it cannot confirm or refute the reported claims of identity impersonation or phishing email activity — that requires the supplementary mail-gateway and identity audit records detailed in Section 8.

2. Reported Incident Background

The narrative motivating this investigation — that Fahad's identity was impersonated to send phishing messages carrying malicious attachments, and that some employees interacted with them — originates from the organizational incident-intake report submitted alongside this case, not from an independent finding within the proxy dataset itself.

This report treats that narrative strictly as a reported claim requiring verification, not as an established fact. Section 6 evaluates each element of it against what the available telemetry can and cannot support.

3. Scope and Evidence Source

Total records analyzed	26 log entries
Sourcetype	bluecoat:proxysg:access:file
User & source host	Fahad3315 → IP 10.55.66.777 (single user/host)
Proxy collector / auth group	10.20.30.40 stc\proxylanusers
Distinct destinations	haveibeenpwned.com, www.virustotal.com, www.zone-h.org, youtube.com (2 variants, excluded)

Background Traffic Excluded: 12 of 26 records are routine YouTube browsing (09:52–09:56, 11:18–11:22) — non-malicious background activity, excluded from the timeline below to keep the analysis focused.

Data Integrity Flag: The recorded source IP (10.55.66.777) is invalid (octet “777” exceeds 255) — likely a masking or export artifact. This reduces confidence in host-level attribution and should be confirmed with log engineering (Section 8).

Inspection constraints: except for zone-h.org, all sessions were TLS-tunneled (port 443). The proxy captured metadata and transfer volumes only — no visibility into payload content, search terms, or scan results.

4. Incident Timeline of Events

Restricted to security-relevant queries and policy-violation events; routine background browsing is excluded per Section 3.

#	Time (UTC+3)	Destination	Status	Bytes In	Bytes Out
1	10:03:34–36	www.virustotal.com	Allowed	2,051	820
2	10:07:37	www.virustotal.com	Allowed	3,152	737
3	10:08:08	www.virustotal.com	Allowed	644,251	4,699
4	11:37:32	haveibeenpwned.com	Allowed	39	231
5	11:38:06–08	www.virustotal.com	Allowed	78	462
6	11:39:24	www.zone-h.org	Blocked	921	521
7	11:39:26	www.zone-h.org/favicon.ico	Blocked	921	411
8	11:42:08	www.virustotal.com	Allowed	2,169	734
9	11:42:19	www.virustotal.com	Allowed	26,571	5,540
10	11:42:38	haveibeenpwned.com	Allowed	106,910	6,999

5. Technical Analysis & Findings

Confidence levels reflect confidence in the behavioral interpretation, not the certainty of the underlying log data — all cited figures are verified as recorded.

Finding 1 — Sequential Cross-Verification Between HIBP and VirusTotal

Interpretation Confidence: **Medium**

Evidence: Timeline entries #4–10 (11:37:32–11:42:38).

Analysis: Within a 5-minute window, the user initiated multiple queries to VirusTotal bracketed by two distinct sessions to HaveIBeenPwned. This is characteristic of an individual checking whether an account or credential was breached while inspecting potentially malicious URLs or files — consistent with either (a) a user actively responding to a perceived security scare, or (b) an unauthorized operator running situational reconnaissance.

Finding 2 — Blocked Cleartext HTTP Requests to a Hacking/Defacement Portal (zone-h.org)

Interpretation Confidence: **Low–Medium**

Evidence: Timeline entries #6–7 (11:39:24–26): two denied requests two seconds apart — the site root and its favicon.ico — both blocked under the Hacking policy category (403 Forbidden).

Analysis: These are the only cleartext HTTP (port 80) requests in the entire dataset — every other session used HTTPS (port 443) — which is why the full request path and referer are visible for this destination alone. The initial request carried `cs_Referer = https://www.google.com/`, indicating navigation via a search-engine result rather than a direct link embedded in an email or document.

Finding 3 — Anomalous VirusTotal Response Volume at 10:08:08

Interpretation Confidence: Low

Evidence: Timeline entry #3: `bytes_in = 644,251` against `bytes_out = 4,699`, versus a 1.9–26.5 KB inbound range for other VirusTotal sessions in the dataset.

Analysis: The large inbound/small outbound asymmetry matches a standard GET-response pattern — retrieving an existing, detailed scan report — rather than uploading a new file for analysis, which would show a comparably large `bytes_out`. Because the session was TLS-encrypted, the specific hash, domain, or IP evaluated cannot be extracted from this log source alone.

Finding 4 — Rapid Service Transitions with a Consistent Interactive Browser Fingerprint

Interpretation Confidence: Low

Evidence: Timeline entries #9–10: a 19-second gap between VirusTotal (11:42:19) and HIBP (11:42:38). All 12 records in the dataset carrying a User-Agent header show an identical signature — standard Chrome 95 on Windows 10 x64 — with no variation anywhere.

Analysis: The short transition interval points to rapid, manual tab-switching by an interactive user reacting to unfolding events. The uniform browser signature, with no automated-scripting indicators (e.g., Python-requests, cURL, headless browsers), corroborates manual operation from a standard workstation rather than a scripted process, though a User-Agent string alone can be spoofed and is not conclusive proof.

6. Incident Hypothesis & Claim Verification

Reported Claim	Investigative Assessment	Telemetry Basis & Rationale
Fahad visited HIBP twice, with VirusTotal activity in between	Confirmed	Directly recorded in telemetry — Timeline entries #4–10 (11:37:32–11:42:38)
Account sent phishing emails impersonating Fahad	Unaddressed by Log Source	Dataset is limited to web-proxy traffic. Validating outbound email requires Exchange/M365 message tracing and SMTP headers
Other employees interacted with malicious messages	Unaddressed by Log Source	Dataset contains telemetry for a single host/account only; other users cannot be measured here
Host or account compromise	Indeterminate	Security-verification activity and one blocked request are circumstantial and equally indicative of legitimate self-investigation

7. Data Limitations

- Scope: 26 records over ~1h50m for a single user; no baseline period for behavioral comparison.
- Payload blindness: TLS-tunneled traffic (port 443) hides queried hashes, URLs, search terms, and scan verdicts.
- Missing telemetry: no EDR, Active Directory / Entra ID sign-in, or mail-gateway data in this source.
- Source IP malformation (10.55.66.777) prevents definitive physical network-jack or DHCP-lease binding.

8. Actionable Recommendations

- Mail Gateway & Message Trace: pull Exchange / M365 Message Trace logs for Fahad3315 on 2021-11-16 — outbound timestamps, sender IP, recipients, SPF/DKIM results.
- Identity & Authentication Audit: review Azure AD / Entra ID sign-in and Unified Audit Logs for unauthorized logins, impossible-travel alerts, or new inbox-forwarding rules.
- Attachment / IOC Correlation: match file hashes or URLs from the reported phishing emails against the VirusTotal session timing in this report (10:03–10:08 and 11:38–11:42).
- Log Pipeline Remediation: escalate the 10.55.66.777 octet anomaly to SIEM/log engineering to confirm root cause.

In summary, this telemetry substantiates that user Fahad3315 was actively conducting security-checking procedures within a tight timeframe, alongside an isolated, blocked attempt to reach an external defacement database via a search-engine result. These artifacts establish investigative activity on the endpoint but do not, on their own, confirm the reported narrative in Section 2; formal attribution should remain bounded until the correlations above are completed.